

A 3D CAD rendering of a large, complex scientific instrument, the Hall A Moller Polarimeter 12 GeV Upgrade. The device is composed of several large, rectangular and cylindrical components mounted on a robust, multi-legged metal frame. The frame features a series of X-shaped bracing structures for stability. The components are arranged in a line, with various pipes, cables, and smaller mechanical parts visible. The entire assembly is shown in a light blue color against a white background.

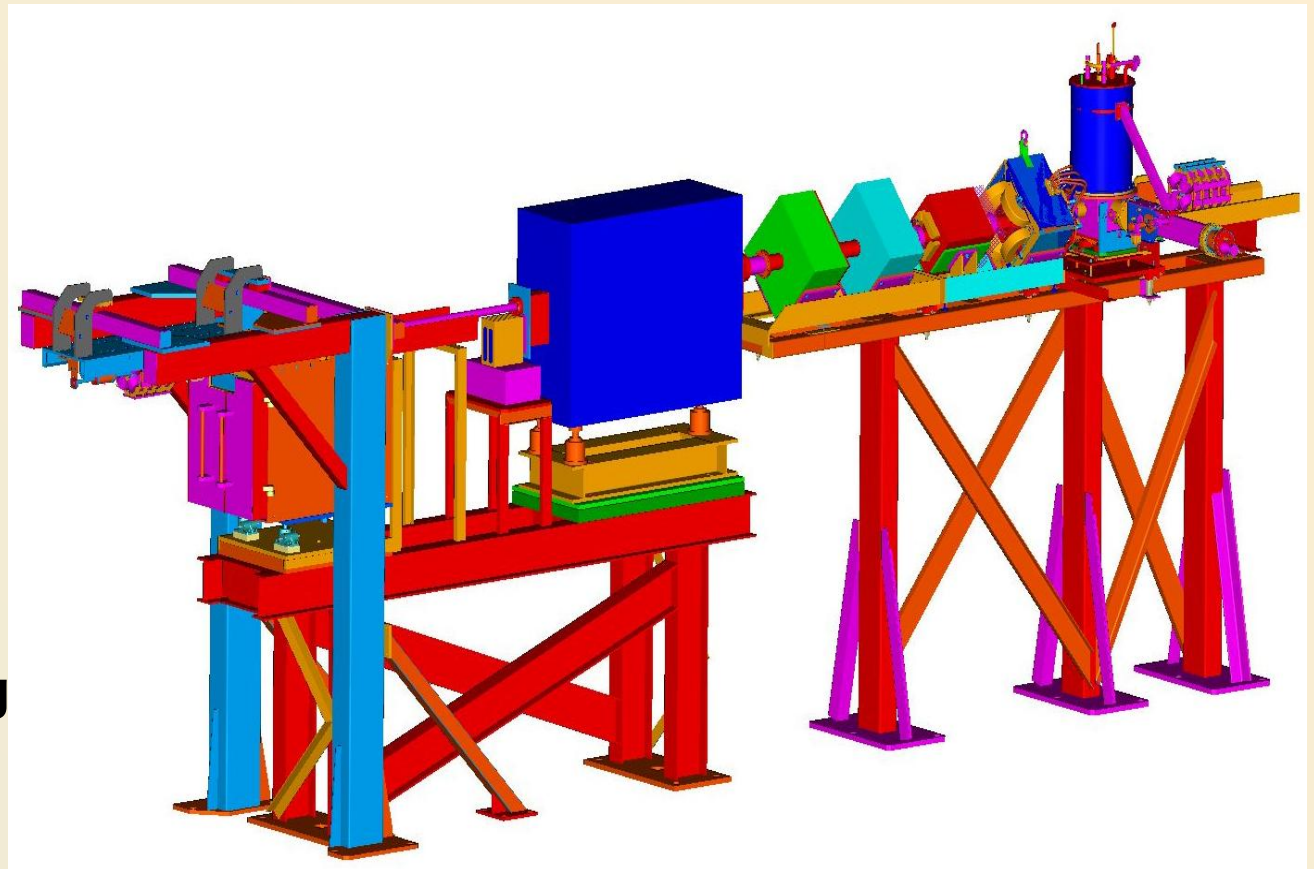
Hall A Moller Polarimeter 12 GeV Upgrade

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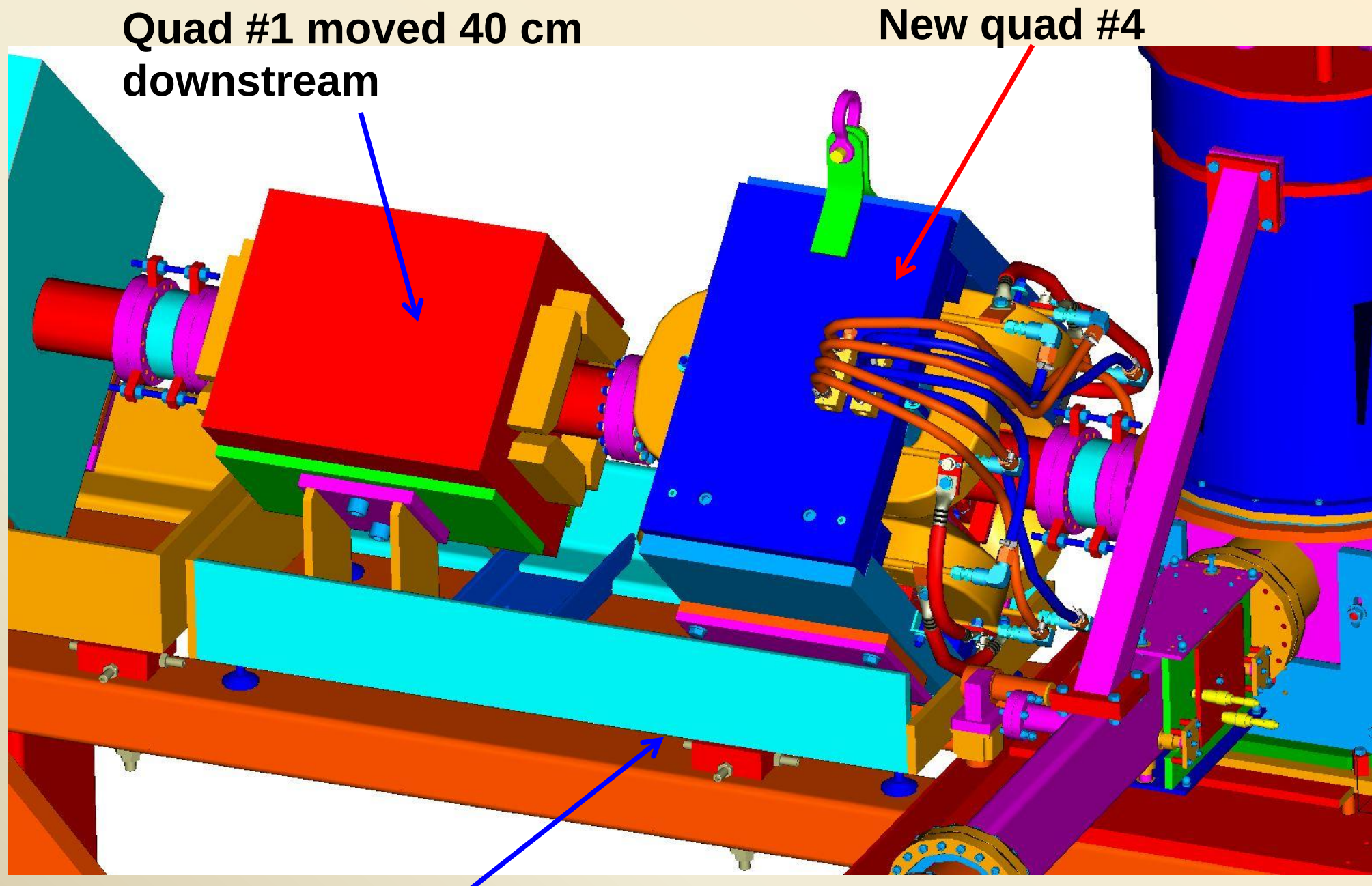
Hall A Collaboration Meeting, June 2012

Hall A Moller Polarimeter for 12 GeV

- Add the 4th quadrupole magnet to cover energy range 0.8~11GeV
- Add a shielding pipe to the dipole magnet
- Lift up the detector shielding box by 10cm Møller electrons bending angle $10^\circ \rightarrow 7.3^\circ$ to use the same dipole power supply
- Rearrange the beam line shielding downstream of the dipole for 7.3° angle
- New correctors and BPM downstream of the Møller dipole
- New detector shielding house



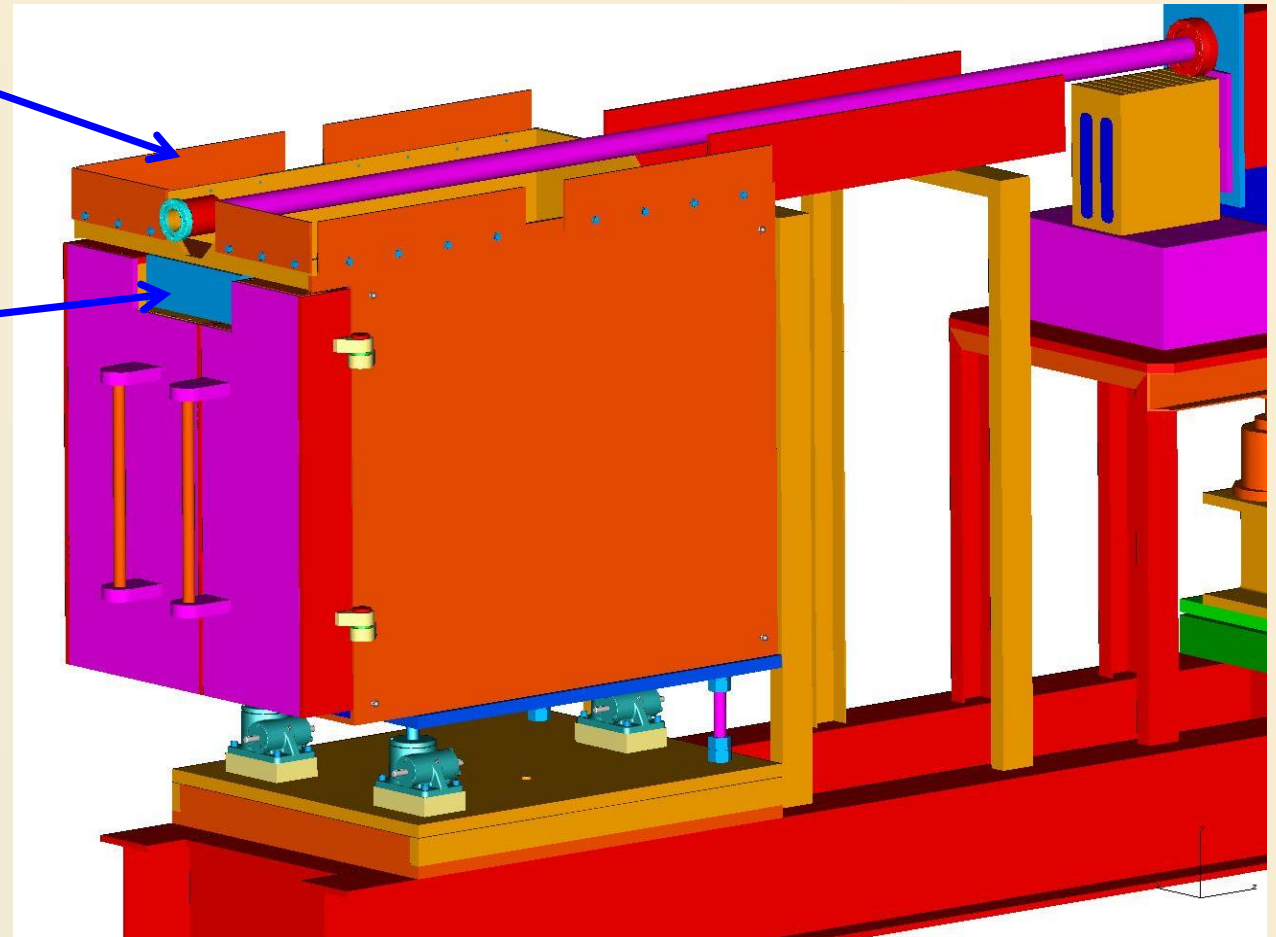
Changes to the Quadrupole Magnets



New girder for Q1 & Q4

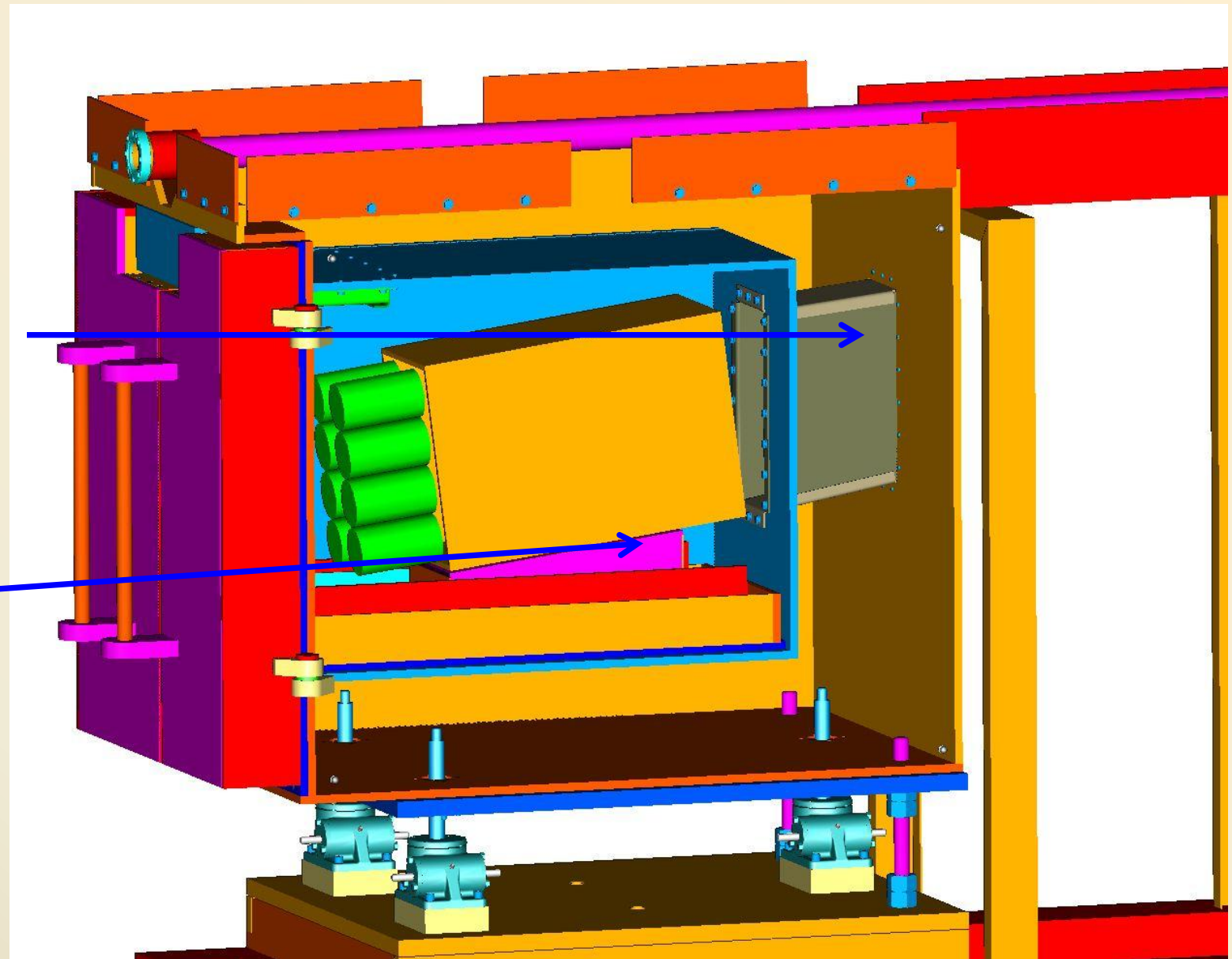
Changes to the Detector Shielding Box

- Top of outer box extended to safely contain lead shielding bricks.
- Clearance in doors for upside down girder components.
- Detector shielding box lifted 7 cm above 6 GeV position.
- Designed to be lifted 10 cm above 6 GeV position if needed.



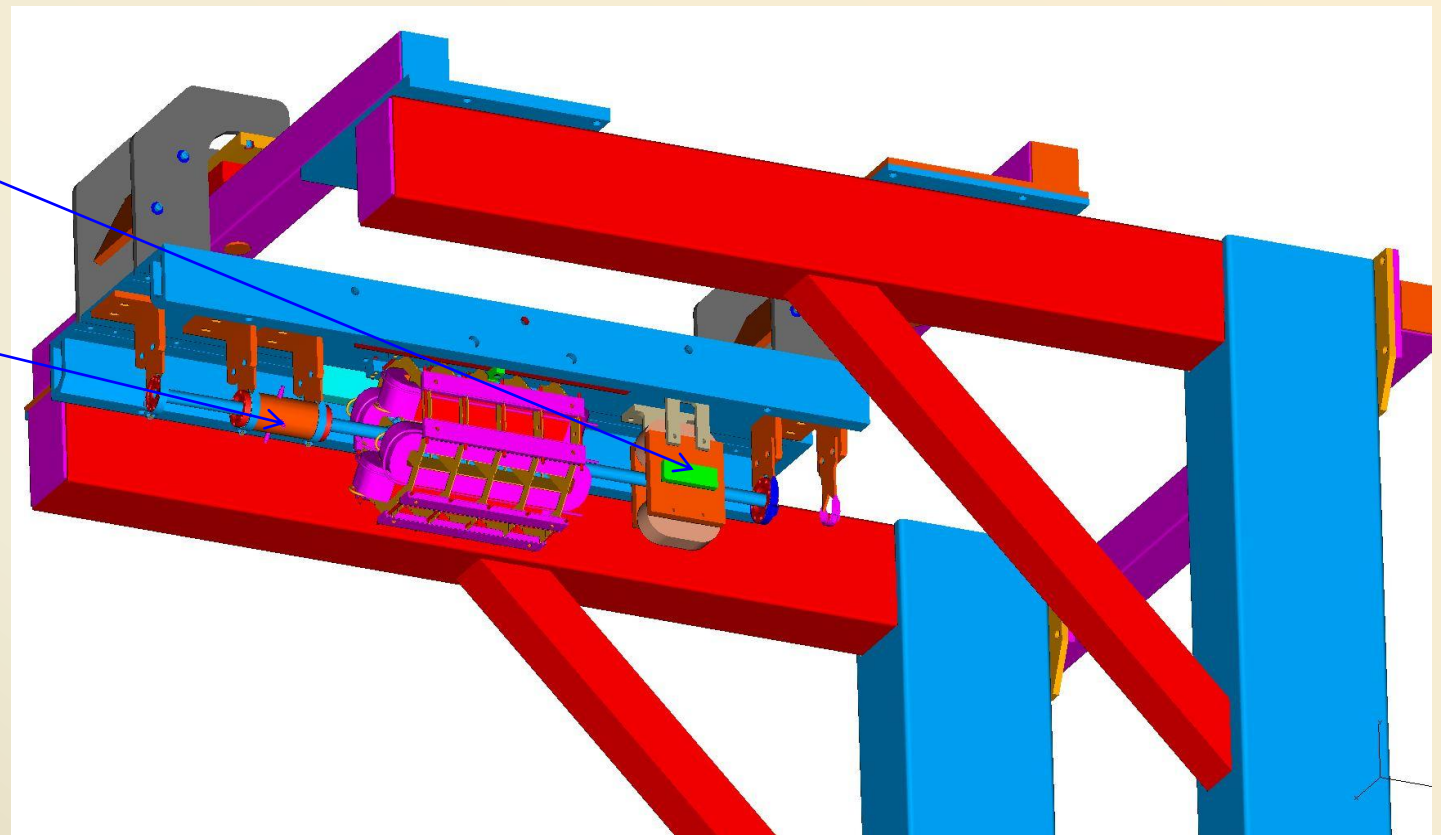
Internal Changes to the Detector Shielding Box

- Added a sheet metal enclosure around the opening of the box.
- Changed the mounting angle of the detector package from 10 deg to 7.3 deg.



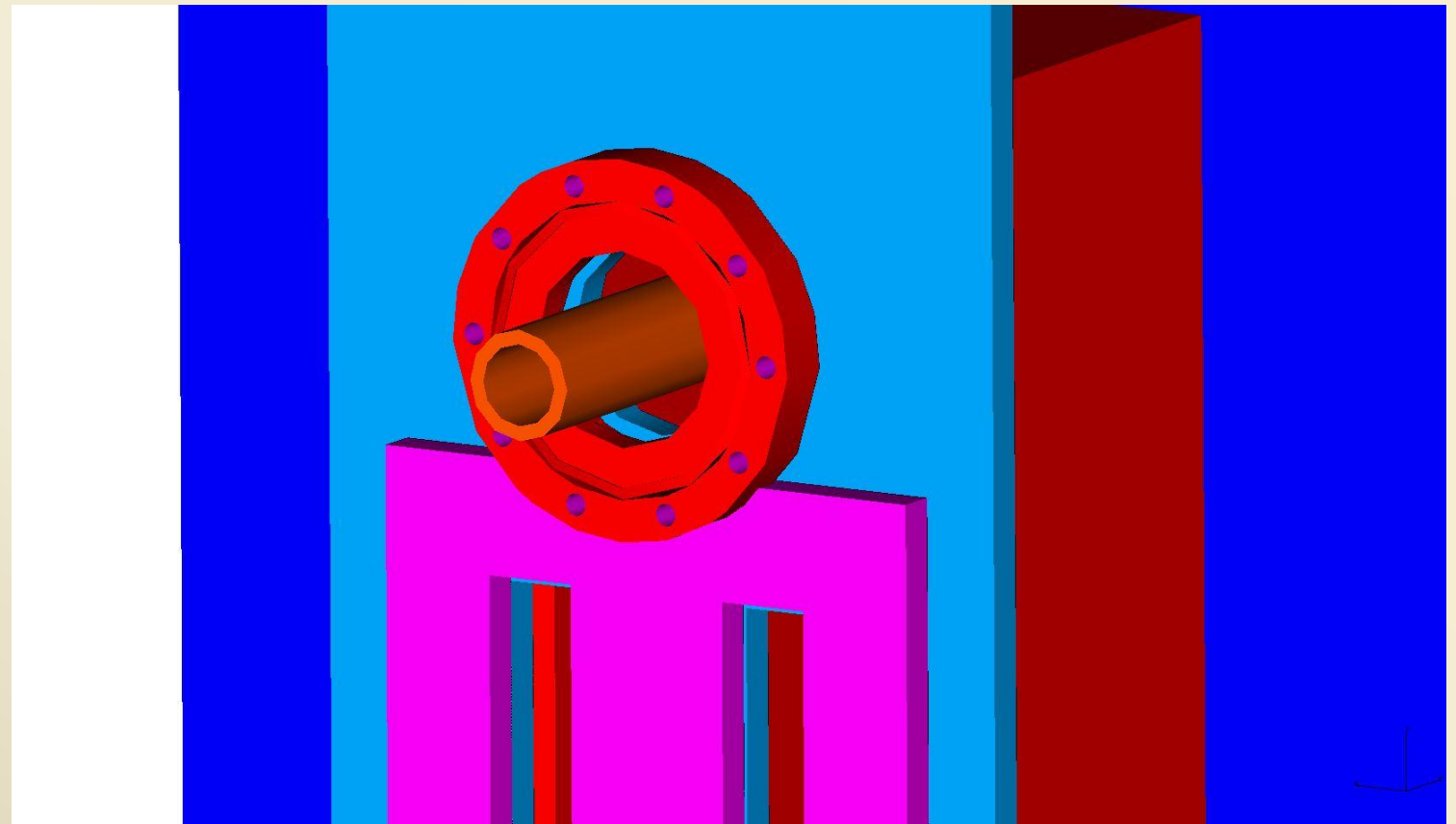
Changes to the Upside Down Girder

- Removed one quad
- Move the remaining quad to a central location
- Added Vertical corrector magnet
- Added a BPM



Changes to the Dipole Magnet

- Beam line Shielding pipe inserted into dipole magnet.
- Rearrange for 7.3 degrees
- Overall length of the pipe is 212.4 cm.



Electronics Upgrades

1) Increase detector bandwidth up to 200MHz

- Programmable Logical Unit(PLU) module LeCroy-2365, bandwidth <75 MHz, CAMAC (**out of market now**)
 ← replaced by PLU module based on CAEN V1495 board (bandwidth 200MHz, VME, FPGA)
- Discriminator Ortec-TD8000, input rate <150 MHz, CAMAC
 ← replaced by P/S 706 (300 MHz, NIM), modified for remote threshold setup with DAC type of VMIC4140

2) Reduce readout time from ADC and TDC modules

- ADC LeCroy 2249A, 12 channels, CAMAC
 ← replaced by QDC CAEN V792 (32 channels, VME)
- TDC LeCroy 2229, CAMAC
 ← replaced by TDC V1190B (64 channels, 0.1ns, VME)

3) We also consider to implement some new triggers unit (for ADC, TDC trigger and gate).

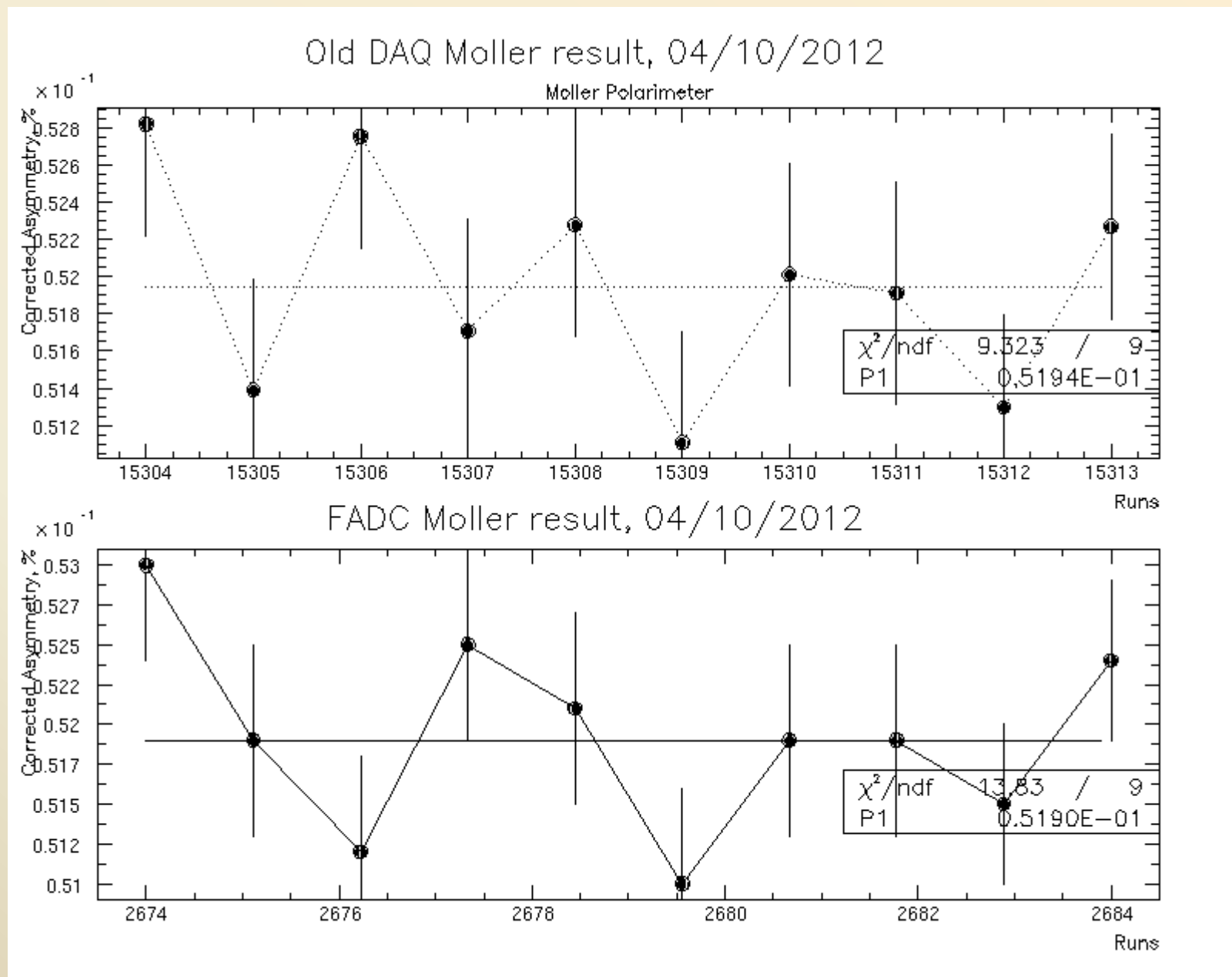
Electronics Upgrades Status

1. Some electronic modules have been installed to VME and NIM crates for testing.
2. Firmware for PLU module is under developing and testing.
3. Software:
 - Library to control DAC for remote threshold setup is ready.
 - Library for configuring and controlling of PLU module is ready (Logical Unit and Scalers).
 - Readout list for TDC and PLU Scalers is ready.
 - Coda for testing TDC and PLU modules is running.

ToDo List.

- Code to analyzing Data Trigger of FADC DAQ
- PLU Firmware optimization
- Continue testing of PLU+TDC+QDC with CODA and Analyzer (modification from old DAQ).
- Migrate both old DAQ and FADC DAQ to CODA 3 (configuration files, mSQL DB, CRL code)

Discrepancy between old and FADC DAQs



Discrepancy is about 0.1-0.5%

Systematic Errors for Current System

Variable	"Low" field	"High" field 3.9T
<i>Target polarization</i>	1.5%	0.35%
<i>Analyzing power</i>	0.3%	0.3%
<i>Levchuk effect</i>	0.2%	0.3%
<i>Target temperature</i>	0.02%	0.02%
<i>Dead time</i>	0.3%	0.3%
<i>Background</i>	0.3%	0.3%
<i>High beam current</i>	0.2%	0.2%
<i>Others</i>	0.5%	0.5%
<i>Total</i>	1.7%	0.9%

Summary

- **Current Moller polarimeter working very well. Discrepancy between old and FADC DAQ is 0.1-0.5%.**
- **The systematic uncertainty of moller measurement with warm target is ~1.7% and 1% for cold target.**
- **Engineering designs are almost finished. Most parts have been delivered to site. New Q4 has been ordered.**
- **Finished Q1 mapping. The understanding of this magnet is much improved.**
- **Electronics module test is in progress.**
- **Software upgrade is going well.**